

BHAVANA DANDU

AI Engineer

(720) 676-3590 | bhavanadandu01@gmail.com | [LinkedIn](#)



PROFESSIONAL SUMMARY:

AI Engineer with **6+ years** of experience building and deploying Machine Learning, Generative AI, and intelligent automation solutions across financial services, predictive analytics, and enterprise environments. Experienced in developing production-grade LLM applications, Retrieval-Augmented Generation (RAG) platforms, multi-agent AI workflows, and predictive analytics solutions that transform complex information into actionable business insights. Skilled in AI engineering, cloud technologies, MLOps, data engineering, and enterprise-scale solution delivery using AWS, Python, and modern AI frameworks. AWS, Azure, and Databricks certified professional with a strong background in building scalable, reliable, and business-focused AI systems.

TECHNICAL SKILLS:

Category	Technologies
Generative AI & LLM	GPT-4, OpenAI API, AWS Bedrock, Lang Chain, LangGraph, Llama Index, Retrieval-Augmented Generation (RAG), Prompt Engineering, Crew AI, AutoGen, Hugging Face Transformers, Semantic Search, Vector Databases (FAISS, Pinecone), Agentic AI, Multi-Agent Systems
Machine Learning	Scikit-learn, XGBoost, Random Forest, Logistic Regression, ARIMA, Prophet, NLP, Anomaly Detection, Feature Engineering, Model Evaluation, Statistical Analysis
Cloud, MLOps & Deployment	AWS (S3, EC2, Lambda, SageMaker, Bedrock, Glue, CloudWatch), Databricks, Docker, Kubernetes, MLflow, Fast API, Flask, REST APIs, CI/CD Pipelines, Model Monitoring, Drift Detection
Data Engineering & Programming	Python, SQL, Pandas, NumPy, Apache Spark, PySpark, Apache Airflow, ETL Pipelines, Data Transformation, PostgreSQL, MySQL, Git, Linux
Visualization & Analytics	Tableau, Power BI, KPI Dashboards, RMSE, MAE, Precision, Recall, F1-Score, ROC-AUC
Domain Expertise	Financial Analytics, Wealth Management, Market Intelligence, Predictive Analytics, Credit Risk Modeling, Fraud Detection, Enterprise Automation, Intelligent Advisory Systems, Customer Analytics, Business Forecasting

PROFESSIONAL EXPERIENCE:

Charles Schwab | Austin, TX

January 2025 – Present

AI Engineer

Project: Generative AI Wealth Management & Intelligent Financial Advisory Platform

Roles and Responsibilities:

- Developed production-grade Generative AI and Retrieval-Augmented Generation (RAG) applications using GPT-4, OpenAI APIs, Lang Chain, AWS Bedrock, and vector search technologies to support wealth management and financial advisory workflows.
- Built scalable AI pipelines integrating financial research documents, SEC filings, investment reports, and enterprise knowledge assets using semantic search, FAISS, and Pinecone, improving information retrieval accuracy and advisor productivity.
- Designed and implemented agent-based AI workflows for document retrieval, compliance validation, financial analysis, and advisor support using Crew AI and AutoGen, reducing manual review effort by 40%.
- Applied prompt engineering, retrieval optimization, and structured output validation techniques to improve response relevance by 25% and enhance reliability across financial advisory use cases.
- Developed and deployed AI microservices using Fast API, Docker, Kubernetes, AWS SageMaker, Lambda, S3, and Glue, supporting enterprise-scale AI workloads and business applications.
- Implemented MLOps practices using MLflow, CloudWatch, model monitoring, drift detection, and evaluation frameworks, reducing model deployment and validation effort by 30%.
- Collaborated with 10+ cross-functional stakeholders including business teams, analysts, compliance teams, and engineers to deliver secure, scalable, and compliant AI solutions while improving operational efficiency by 25%.

Data Scientist**Project:** AI-Driven Financial Market Intelligence & Predictive Analytics Platform**Roles and Responsibilities:**

- Built scalable ETL and analytics pipelines using Python, Pandas, SQL, and AWS services to process 5M+ financial data records from enterprise and third-party sources.
- Performed exploratory data analysis, statistical analysis, and hypothesis testing to identify market trends, financial risks, and key business performance drivers.
- Developed machine learning and forecasting models using XGBoost, Random Forest, Logistic Regression, ARIMA, and Prophet for credit risk analysis and revenue forecasting, improving forecasting accuracy by 22%.
- Designed anomaly detection frameworks, feature engineering workflows, and hyperparameter tuning processes using Scikit-learn and GridSearchCV, improving model reliability and predictive performance.
- Created 10+ interactive Tableau and Power BI dashboards to visualize KPIs, forecasting insights, and model performance metrics for business stakeholders and leadership teams.
- Leveraged AWS services including S3, EC2, and SageMaker for scalable model training and deployment across analytics use cases.
- Automated reporting and analytics workflows, reducing manual reporting effort by 65% and improving turnaround time for business users.
- Collaborated with 10+ stakeholders including business analysts, data engineers, and leadership teams to deliver predictive analytics and business intelligence solutions.

Infocusp | India

March 2019 – April 2022

Python Developer**Project:** Intelligent Data Processing & Enterprise Automation Platform**Roles and Responsibilities:**

- Developed modular ETL pipelines in Python using Pandas and NumPy to extract, clean, validate, and transform 2M+ business records from multiple enterprise data sources.
- Build reusable automation framework for data validation, deduplication, reporting, business rule enforcement, and enterprise data processing through REST API integrations.
- Developed and optimized SQL queries, joins, and stored procedures across PostgreSQL and MySQL databases to support large-scale data processing and reporting workflows.
- Automated Excel-based reporting and batch-processing workflows, reducing manual effort by 75% and improving operational efficiency across business teams.
- Implemented logging, exception handling, monitoring, and performance optimization techniques, reducing workflow execution time by 35% and improving reliability.
- Created analytics-ready datasets and reporting layers supporting 10+ Tableau dashboards for KPI tracking, operational insights, and executive reporting.
- Supported deployment and application maintenance using Docker, Git, Jenkins, and CI/CD pipelines while collaborating with 8+ cross-functional stakeholders to deliver scalable automation solutions.

CERTIFICATIONS:

- AWS Certified AI Practitioner
- AWS Certified Machine Learning Engineer - Associate
- Microsoft Certified - Azure AI Engineer Associate
- Databricks Certified Generative AI Engineer Associate

EDUCATION:**Master's** in Business Analytics - University of Colorado Denver, CO